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Video smoke separation and detection via sparse representation

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ABSTRACT

Most vision-based object detection methods rely heavily on feature extraction. For opaque objects, features extracted directly from the source image in motion area or on block level can provide reliable information for detection. But for smoke detection, due to its transparency, variable thickness and scattering, conventional features may not be effective. In this paper, smoke components are represented in sparse coefficients over learned smoke dictionary on block level. The process is mathematically modeled by considering the observed block image as a linear combination of smoke and background components. We propose a novel regularization and variable selection approach with elastic net solutions and discriminative constraints. The discriminative feature with respect to the sparse coefficients is selected for smoke classification and detection. Experimental results on synthesized and real video data sets show that the proposed approach outperforms conventional methods and has a significant performance enhancement.

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1. Introduction

Smoke detection should be performed in the early stage before the onset or eruption of a fire event. Conventional points smoke detection are widely used in indoors environments. They typically detect the presence of certain particles generated by smoke and fire using ionization or photoelectric detectors. However, fire alarms will not be issued unless particles reach the sensors to activate them. Smoke temperature or infrared detector require a close proximity. To cover all desired areas, multiple detectors are needed which is not cost-effective. These drawbacks limit the detection performance of these detectors in outdoor environments. Moreover, these detectors cannot provide detailed information about the location of smoke, its scale and intensity [1–3].

Recent advances in computer vision make vision-based smoke detection have the potential to overcome the above drawbacks. It can be used in both enclosed and open spaces and can provide broad and large enough views of smoke and fire with locations and other key indicators. To the best of our knowledge, the success of vision-based smoke detection techniques depends highly on identification of reliable features that can characterize smoke. Despite the efforts in [1,2,4], it is still difficult to quantify the visual characteristics of smoke.

Herein lies the challenges, the background covered by smoke may be included in the feature being extracted, so that direct feature extraction may not work well for smoke detection. One needs to account for the varying degree of transparency. Fig. 1 gives the possible effects in the case of light smoke. The images shown in Fig. 1(a) illustrates the light smoke covered with background in current input video frame. Fig. 1(b) illustrates the presence of residual background information when a conventional video-based background subtraction method (robust orthonormal subspace learning (ROSL) [5]) is used. It is summarized that when backgrounds are covered by opaque objects, the features can be extracted and used directly without the impacts of background. However, smoke may be transparent and it may act as the scattering medium like the fog/haze [6]. The changing thickness of smoke also have different levels of impact on smoke detection.

All these problems described above motivate us to seek an solution to address smoke separation and reliable feature extraction from pure smoke for vision-based smoke detection. Tian et al. [7] proposed three smoke separation methods based on different models for smoke components: (1) Local smoothness model: it is based on the observation that pure smoke image in a small-sized block is generally smooth. In other words, pixels in a pure image block are spatially close and visually similar in intensities or colors. (2) Principal component model: while the local smoothness model would fail in distinguishing smoke from other opaque objects whose surfaces are also smooth, or it would be ineffective when background is also smooth or piece-wise smooth, it is considered that pure smoke images, being similar in overall textural

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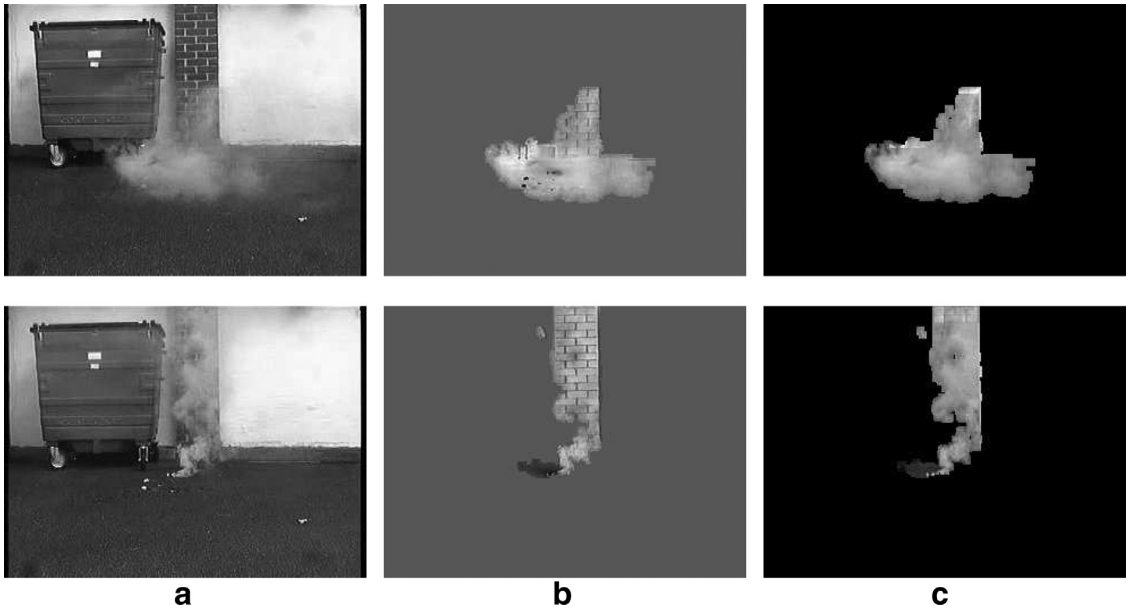

Fig. 1. (a) Current input frame; (b) Foreground; (c) Separated Smoke.

configuration, are likely to lie in a low-dimensional subspace, principal component analysis (PCA) is employed to locate the subspace of pure smoke images. (3) Sparse representation model: a single linear subspace obtained by PCA may not be sufficient to describe all possible variations of smoke components since they may lie in multiple low-dimensional subspaces. According to the theory of sparse representation [8], if sample smoke images can be collected or generated to capture the distribution of pure smoke images, it is expected that any specific pure smoke image would have a sparse representation with respect to these samples. Such collection of samples represent a dictionary and each sample in the dictionary is typically referred to as a basis. While the sparse representation performs the best in smoke separation among these three models, the extracted sparse coefficients are found to be insufficient to achieve satisfactory results in smoke detection [7]. Further study is needed to understand how and in what way can the extracted sparse coefficients characterize smoke. In the case that the number of predictor p is larger than the number of variables n [9], the lasso [7], as the optimization solution, can select at most n variables. If there is a group of variables among which the pairwise correlations are high, then the lasso tends to select only one variable from the group and does not care which one is selected. To address these issues, elastic net, a generalization of the lasso, has been shown to be a valuable tool for model fitting and feature extraction [9]. In this paper, we improve the sparse representation method [7] and elastic net is used there instead of lasso optimization. The smoke separation performance is shown in Fig. 1(c), which depicts only smoke components without background components, the features from separated pure smoke component are selected for smoke detection. Additionally, inspired by Discriminative K-SVD (DKSVD) [10], in classification tasks based on sparse representation, the discriminative ability of sparse coefficients is essential except the reconstruction ability, hence improvements are also made for separation and detection by adding the discriminative constraint.

Following the introduction, the reviews of related work about vision-based image separation from image or video and smoke detection methods are introduced in Section 2. The smoke separation method is presented in Section 3. The proposed method for smoke detection is described in details in Section 4. Performance evaluations on smoke separation and detection are given in Section 5 along with the discussion. The paper is concluded in Section 6.

2. Related works

Here we review some relevant image-based or video-based image separation methods. Some representative vision-based smoke detection methods are also mentioned.

2.1. Image separation from image or video

Image separation methods can be categorized as image-based and video-based separation. For image-based separation, a well-known corresponding research topic called blind source separation (BSS) has been investigated [11,12]. It consists of separating a set of unknown signals or image information from the linear mixtures of them. It is also widely used in the “cocktail party” problem. Many techniques have been proposed to solve BSS, such as independent component analysis (ICA) [13] methods based on the assumption of mutual independence of the sources, maximum likelihood (ML) [14] method by assuming a factorized form for the joint source distribution. Bayesian method is another approach for BSS. It is a natural and flexible way to account for prior knowledge. Models under the form of suitable priors are used for representing the local smoothness and enforcing independence of original images [15,16]. The ICA or ML methods for separation require different observations like illuminations, for smoke detection, but only one mixture observation of smoke and background is available. In addition, Bayesian approach cannot be used in smoke detection while either smoke or background in local area may be smooth, and there is no simple prior knowledge that can separate smoke and background independently. Hence these methods are of limited use for separating smoke components based on single image. Another representative approach for image separation from single image is the decomposition of images into piece-wise smoothness (cartoon) and texture layers reported in [17–19]. A combination of sparse representation and a variational approach is proposed by Starck et al. [19]. However, this method is not useful for smoke separation because piecewise smoothness or texture property can be either from smoke or background or the mixture of them. Tian

et al. [6] propose a similarity-based smoke separation method. Assume that pixels which are visually similar and spatially close have similar blending parameter between smoke and background. The pixels classified as smoke or non-smoke with high scores will be regarded as the reliable known pixels. This method depends highly on previous smoke detection results and the similarity function is not sufficient to describe the details of smoke too.

In video-based image separation, original video frame images cannot provide reliable information of smoke due to the interference of background blending with smoke. Great efforts have been made for foreground separation or background subtraction. A well-known non-parametric background model called ViBe is proposed in [20,21], pixel values in background are updated by a random scheme, and updated pixels can diffuse their current pixel into neighboring pixels using another random selection method. This method uses a similar sample-based background model that relies on color. Gaussian Mixture Model (GMM) [22] is another well-known method to obtain an approximation of background. Afterwards, the pixel-based methods are replaced by large amounts of low-rank and sparse decomposition methods such as robust principal component analysis (RPCA) [23], robust orthonormal subspace learning (ROSL) [5], principal component pursuit (PCP) [24], robust pca via gradient descent (RPCA-GD) [25], SemiSoft GoDec (SGoDec) [26], the success of these algorithms depends on the assumption that background shows absolute low-rank property without local illumination changes while the moving objects are sparse in observed matrix. Smoke usually occupies locally in the scene and will generate airlight, a mechanism that causes the atmosphere (i.e. smoke here) to behave like a source of light through scattering [7], and background will blend with sparse foreground such that these methods may fail in decomposition of background and smoke components. Three variants of smoke separation models are proposed in [7] such as local smoothness model, principal component model and sparse representation model. All these models provide efficient ways for smoke detection, comparison and the limitation of these methods are shown as well in this literature.

We also refer to some researches on rain streaks removal from single image or video due to its similarity to smoke separation with transparent foregrounds. Zhang et al. [27] and Chen et al. [28] follow the same idea that the rain streaks in block have similar patterns and can be described into low-rank model, such that the rain streaks can be removed from image by capturing the spatio-temporal correlation. Convolutional sparse coding (CSC) [29,30] or multiscale convolutional sparse coding (MS-CSC) [31] method learn the convolutional dictionary of rain streak with its sparsity. Nevertheless, smoke with different thickness or partial coverage may result in variant patterns, and different from the task of separation, these methods aim at filtering out the rain streak out but not separate it from background.

Despite the limits of these methods for smoke separation, they also contribute many thoughts for solving the dilemma between smoke separation and background modeling in an efficient way, and our purpose is smoke detection using features extracted from separated pure smoke components.

2.2. Vision-based smoke detection

Existing methods for smoke detection mainly depend on analyzing statistic or dynamic features in different mechanisms. Frequent use of color, texture, edge or contour-based features along with the use of dynamic features are reported in great quantity of literature datum.

Grayish color of smoke makes it a common feature used in smoke detection [2,32–35], despite different description of these color features, they share the similar rules that smoke pixel values from RGB three channels are equal or similar. Except the analysis on RGB channels, an analysis in Lab color space is also performed in [36] for smoke saliency enhancement. The color-based smoke detection method uses transmission estimation with dark channel prior and the final result is refined through soft matting algorithm. Definitely, color features extraction methods are not appreciated in grayscale image detection due to the use of different color channels or dark channel prior.

Edge or texture based methods mainly follow two different ideas, one adopts the features extracted directly from smoke which can characterize the edge or texture of it, histogram orientated gradient (HOG) [37] is a representative edge-based descriptor for smoke detection [38,39], and local binary pattern (LBP) [40] has also been successfully applied to texture classification. These two descriptors work by describing the boundary or local boundary information of smoke in details. Another edge-based detection method relies on the smoke effects for edge or texture. It is well known that edges always have high frequencies while the smoke blurred edges will contribute to the decrease of its high frequency. Therefore, some researches analyze the high frequency information in different direction for smoke detection in wavelet domain [40,41].

The linear decomposition of data using a few elements from a corresponding learned dictionary or data sets lead to state-of-the-art results in recent researches about low-level signal processing tasks and features analysis, such as audio processing [42–44], image denoising [45–47], image classification [48–53], and industrial applications [54–56]. Image can be represented by the learned dictionary with its sparse coefficients, the coefficients act as hidden/latent feature vectors to encode more discriminative information and can be used in image classification [6,57,58]. Tian et al. [6] solve the sparse representation problem using dual dictionaries for smoke and background components respectively. A feature is constructed as a concatenation of the respective sparse coefficients for detection.

As discussed above, linear decomposition of signals using a few atoms of a learned dictionary yield not only sparse representation as a predefined method (like wavelet, curvelet, DCT etc.) but also discriminative information. A well-known dictionary learning method is K-SVD [59]. For a given training samples x, the K-SVD algorithm tries to minimize the representation errors as

$$\arg\min_{D,\alpha} \|x - D\alpha\|_2^2 \quad s.t. \quad \|\alpha_i\|_0 \leqslant T \quad \forall i, \tag{1}$$

where $x = [x_1, x_2, \ldots, x_n] \in \mathbb{R}^{m\times n}$ denote the input training data sets, $D \in \mathbb{R}^{m\times L}$ is the dictionary, and $\alpha \in \mathbb{R}^{L\times n}$ is the coefficients matrix with its columns $\alpha_i \in \mathbb{R}^L$.

The optimization of Eq. (1) proceeds in two stages. The first stage learns the dictionary by computing the singular value decomposition (SVD), and the second stage solves the $\ell_0 - norm$ minimization problem. Sparse representation over dictionary learning for image classification tasks have been reported in many literatures, such as Discriminative K-SVD (DKSVD) [10], Fisher Discrimination Dictionary Learning (FDDL) [60], Label Consistent K-SVD (LCSVD). [61] etc.

3. Smoke separation

3.1. Problem formulation

According to attenuation and airlight models developed for atmospheric scattering [62], by ignoring the multiple scattering, the observed intensity of smoke at pixel y [63] can be expressed as

$$X(y) = B(y)\Omega(y, \Delta z) + S(y, z)(1 - \Omega(y, \Delta z)), \tag{2}$$

where X, B and S account for the input image, background and smoke respectively. The smoke components at different distance

z are considered corresponding to the thickness Δz, while $\Omega \in [0, 1]$ are referred to as blending parameter and vary from different pixels y. Our goal is to estimate the background B, smoke S and blending parameter Ω with the observed X.

Generally, it is assumed that smoke in a small block often share the similar blending parameter, such that the blending parameter in this area is a constant and the smoke image is a linear combination of background and pure smoke with the same blending parameter [6,7,63] Observation and process are focused on block-level images by dividing image into overlapped or non-overlapped blocks. Therefore, the problem is reduced to classifying these small-sized blocks as smoke or non-smoke.

Let $x \in \mathbb{R}^m$ be the observation matrix of an image block with m pixels,

$$x = \omega s + (1-\omega)b + \varepsilon \quad s.t. \omega \in [0, 1], \tag{3}$$

where $\varepsilon \in \mathbb{R}^m$ represents the noise of model, $s \in \mathbb{R}^m$ represents matrix of the pure smoke in x and $b \in \mathbb{R}^m$ represents the background under clear air without smoke coverage. $\omega \in [0, 1]$ denotes the blending parameter. The blending parameters of different blocks may vary from each other for different transparency degree or thickness of each smoke block.

Assume x is acquired by a stationary camera, the background component can be estimated by background modeling techniques such as Gaussian mixture model (GMM) [64], robust principal analysis (RPCA) [23], principal component pursuit (PCP) [24], robust orthonormal subspace learning (ROSL) [5] and so on. In this paper, ROSL is used for background subtraction due to its advantage in fast computation and robust performance [5]. The estimates of ω and s with given x and b can be obtained by minimizing the residual noise:

$$(\omega, s) = \underset{\omega,s}{arg\min} \|x - \omega s - (1-\omega)b\|_2^2 \quad s.t. \omega \in [0, 1]. \tag{4}$$

To obtain a unique and reliable solution of Eq. (4), further constrains are required. Three constraints are introduced in [7] based on different properties of smoke: local smoothness, principal component and sparse representation. Since the sparse representation model is proved to outperform the other twos in smoke separation, we propose an improved sparse representation method here to separate smoke components. Comparison will be carried out in the experiments to validated the effectiveness and improvements of the proposed method.

3.2. Sparse representation for smoke component separation

Sparse Representation with respect to α_s: Since natural images can be represented on a data-driven dictionary as a linear combination of sparse representation [8], it is suggested that any specific smoke image can be represented using sparse coding with respect to the smoke image samples collected for capturing the distribution of pure smoke components.

Consider that it admits a sparse approximation over an overcomplete smoke dictionary $D_s = [d_1, d_2, ..., d_L]$ in $\mathbb{R}^{m\times L}$ $(L \gg m)$ when one can find a linear combination for L columns of D_s that is equal or close to the smoke vector s, the coefficients of the linear combination can be regarded as a sparse coefficient vector $\alpha_s \in \mathbb{R}^L$ in sparse feature space. A pure smoke image is expected to be sparse in D_s and can be reconstructed by:

$$s = D_s \alpha_s. \tag{5}$$

The sparsity of α_s determines that many elements are zero or close to zero. The coefficients vector indicates which atoms and the proportion thereof contribute to the construction of quasi-smoke components. Thus α_s is expected to encode the discriminative information of smoke. This sparse representation model can be expressed as [6,7]

$$\min_{\omega,\alpha_s} \|x - \omega D_s \alpha_s - (1-\omega)b\|_2^2 + \lambda \|\alpha_s\|_1 \quad s.t. \omega \in [0, 1], \tag{6}$$

where $\ell_1 - norm$ is used as the constraint of sparsity instead of $\ell_0 - norm$ since the $\ell_0 - norm$ is non-convex and the solution to it is a NP hard problem.

Fix ω, Eq. (6) with constraint of $\alpha_s$is a classical lasso problem, it is of much success in both continuous shrinkage and automatic variable selection. Many approaches, such as generic QP solvers (e.g., CVX), Chen et al.'s interior point method [65], a modification of least angle regression (LARS) [66], grafting [67] and feature-sign search [68] have been proposed. However, lasso is not effective when the number of predictors is much larger than the number of observations $(L \gg m)$ [9]. It does not perform well in variables selection of feature, and it lacks the ability in revealing grouping information.

Here the elastic-net is adopted instead of feature-sign search to figure out the variables in α_s. Replace Eq. (6) with

$$\begin{aligned} &\min_{\omega,\alpha_s} \|x - \omega D_s \alpha_s - (1-\omega)b\|_2^2 + \lambda_1 \|\alpha_s\|_1 + \lambda_2 \|\alpha_s\|_2^2, \\ &s.t. \omega \in [0, 1] \end{aligned} \tag{7}$$

ω and α_s can be updated by fixing them alternately.

When ω is fixed, Eq. (7) becomes a problem of regularization and variable selection with respect to α_s,

$$\min_{\alpha_s} \|x - \omega D_s \alpha_s - (1-\omega)b\|_2^2 + \lambda_1 \|\alpha_s\|_1 + \lambda_2 \|\alpha_s\|_2^2. \tag{8}$$

To solve for α_s, the naïve elastic net criterion is written as [9]

$$\mathscr{L}(\alpha_s, \lambda_1, \lambda_2, \omega) = \|x - \omega D_s \alpha_s - (1-\omega)b\|_2^2 + \lambda_1 \|\alpha_s\|_1 + \lambda_2 \|\alpha_s\|_2^2, \tag{9}$$

Define the artificial data sets (y^*, D_s^*) from the given data set (x, D_s) and (λ_1, λ_2) as

$$y^* = [x - b + \omega b; 0], D_s^* = \frac{1}{\sqrt{1+\lambda_2}}\left[\omega D_s; \sqrt{\lambda_2} I_L\right]. \tag{10}$$

where $I_L \in \mathbb{R}^{L\times L}$ is an identity matrix. Let $\gamma = \frac{\lambda_1}{\sqrt{1+\lambda_2}}$ and $\alpha_s^* = \sqrt{1+\lambda_2}\alpha_s$, Eq. (9) can be rewritten as

$$\mathscr{L}(\alpha_s, \gamma, \omega) = \mathscr{L}(\alpha_s^*, \gamma, \omega) = \|y^* - D_s^* \alpha_s^*\|_2^2 + \gamma \|\alpha_s^*\|_1, \tag{11}$$

such that the solution to Eq. (8) becomes

$$\widehat{\alpha}_s^* = \underset{\alpha_s^*}{\arg\min} \mathscr{L}\{(\alpha_s^*, \gamma, \omega)\}. \tag{12}$$

This procedure simplifies the naïve elastic net problem as a lasso type problem on the augmented data set (λ_1, λ_2)

$$\widehat{\alpha}_s^* = \underset{\alpha_s^*}{\arg\min} \|y^* - D_s^* \alpha_s^*\|_2^2 + \frac{\lambda_1}{\sqrt{1+\lambda_2}} \|\alpha_s^*\|_1, \tag{13}$$

The elastic net estimation of $\widehat{\alpha}_s$ is defined as

$$\widehat{\alpha}_{s(en)} = \sqrt{1+\lambda_2}\widehat{\alpha}_s^*. \tag{14}$$

Recall the naïve elastic net estimation that $\widehat{\alpha}_{s(nen)} = \frac{1}{\sqrt{1+\lambda_2}}\widehat{\alpha}_s^*$, and we have

$$\widehat{\alpha}_{s(en)} = (1+\lambda_2)\widehat{\alpha}_{s(nen)}. \tag{15}$$

Referred to [9], note that the elastic net estimates $\widehat{\alpha}_s$ can be given by

$$\begin{aligned} \widehat{\alpha}_s = \underset{\alpha_s}{\arg\min}\, &{\alpha_s}^T \left(\frac{\omega^2 {D_s}^T D_s + \lambda_2 I_L}{1+\lambda_2}\right)\alpha_s \\ &-2\omega(x - b + \omega b)^T D_s \alpha_s + \lambda_1 \|\alpha_s\| \end{aligned} \tag{16}$$

while the lasso estimates can be expressed as

$$\alpha_{s(lasso)} = \arg\min_{\alpha_s} {\alpha_s}^T\left(\omega^2 {D_s}^T D_s\right)\alpha_s - 2\omega(x-b+\omega b)^T D_s\alpha_s + \lambda_1\|\alpha_s\|, \quad (17)$$

For a fixed λ_2, the elastic net problem can be solved by following the idea of lasso solution. It brings extra flexibility with the cost of an additional design parameter λ_2 which can be obtained by a modified LARS algorithm called LARS-en [9]. It creates the entire elastic net solution path step by step and arrive at the best group tunning parameters (λ_1, λ_2) with the smallest objective error.

Extension-discriminative sparse coefficients: Referred to discriminative dictionary learning method for image classification, besides the reconstruction ability of sparse representation over learned dictionary, a powerful discriminative ability is also important for classification [69–71]. The discriminative ability for sparse coefficients α_s means that the corresponding smoke dictionary D_s has good representation ability to the smoke class images while the non-smoke dictionary D_b has poor representation ability for them. In other words, any images reconstructed using sparse coefficients over learned overcomplete smoke dictionary D_s should be 'penalized' by the non-smoke dictionary D_b, so that the α_s represented over D_b results in small value, this mechanism guarantees that atoms in α_s are of different proportion to the construction of the smoke and non-smoke components. Let D_b represent the learned non-smoke dictionary, the discriminative model is defined by extending Eq. (7) as:

$$\min_{\omega,\alpha_s} \|x-\omega D_s\alpha_s-(1-\omega)b\|_2^2 + \|D_b\alpha_s\|_2^2 + \lambda_1\|\alpha_s\|_1 + \lambda_2\|\alpha_s\|_2^2, \quad s.t.\omega\in[0,1]. \quad (18)$$

The discriminative ability of the features come from the minimization of the value of ℓ_2-norm $\|D_b\alpha_s\|_2^2$. The coefficient vector α_s indicates which atoms and the proportion thereof that contribute to the construction of the smoke components, while atoms in a discriminative coefficient vector of smoke should contribute to non-smoke dictionary based construction. The solution to Eq. (18) can be simply issued by adding the extended constraint into Eq. (10) as follows:

$$y^* = [x-b+\omega b; 0; 0], D_s^* = \frac{1}{\sqrt{1+\lambda_2}}\left[\omega D_s; D_b; \sqrt{\lambda_2}I_L\right]. \quad (19)$$

then follow the same process from Eq. (11) to Eq. (18).

Update blending parameters ω: Let $\widehat{\omega}$ be the current estimates of ω with a fixed α_s, Eq. (7) is a quadratic function of ω and the minimization of this function with respect to ω is given by:

$$\omega^* = \frac{(b-D_s\alpha_s)^T(x-b)}{(b-D_s\alpha_s)^T(D_s\alpha_s-b)}. \quad (20)$$

Since $\omega\in[0,1]$, $\widehat{\omega}$ can be achieved by posting the constraint:

$$\widehat{\omega} = \begin{cases} 0, \omega^* \leqslant 0 \\ \omega^*, \omega^* \in (0,1) \\ 1, \omega^* \geqslant 1 \end{cases} \quad (21)$$

The proposed algorithm is summarized in Algorithm 1.

Algorithm 1 Smoke Separation via Joint Sparse Coding.

Input: Block image $x\in\mathbb{R}^m$; Background $b\in\mathbb{R}^m$; Smoke Dictionary $D_s\in\mathbb{R}^{m\times L}$; Regularization parameters λ_1, λ_2.
Output: α_s, ω, s

```
Initialize ω_0 = 0.5
while not converged do
    Update α_s of the sparse coding problem
    min_{α_s} ||x − ωD_sα_s − (1 − ω)b||_2^2 + ||D_bα_s||_2^2 + λ_1||α_s||_1 +
    λ_2||α_s||_2^2, s.t.ω ∈ [0, 1]
    by following the solution in section 3.2
    Compute ω using Eq. (20) and Eq. (21);
end while
return s = D_sα_s, ω.
```

4. Smoke detection

Fig. 2 depicts the proposed video smoke detection framework. Given a video, background modeling is performed, the current image sequence and its estimated background are divided into overlapped or non-overlapped small blocks x and b. Smoke separation techniques is then adopted to estimate the smoke components s represented by $s = D_s\alpha_s$ and blending parameters ω on block level. Too small blending parameters are determined as that the block x is not covered by smoke or anything else and are classified as non-smoke. We set a small threshold T_ω to filter out these non-smoke blocks when $\omega < T_\omega$. When $\omega \geqslant T_\omega$, further classification is conducted on separated component s or sparse coefficients α_s. Since LBP feature extracted from separated smoke components s is successfully applied to smoke detection [7], it is selected as another feature compared with the proposed sparse coefficients vector in the following experiments. In this paper, the sparse coefficients vector α_s is input to an SVM classifier as feature. Decision is made on whether this feature can contribute to smoke over learned smoke dictionary D_s.

5. Performance evaluation

To evaluate the performance of the proposed method for smoke separation and detection, In this section, the following experiments are conducted along with the discussion of corresponding experimental results:

(1) Smoke separation on synthesized image samples combined using pure smoke and non-smoke with different blending parameters in different changing precisions.
(2) Visual assessment of smoke separation performance on block images and video frame images.
(3) Smoke classification on synthesized image samples using known background covered by light or heavy smoke by controlling the values of blending parameters. Different sparse representation methods over dictionary learning are compared.
(4) Smoke detection on real video images.

Before presenting the experiments, we briefly introduce the question of choosing the parameters in our formulation and the data sets used in this paper.

5.1. Choice of parameters

As discussed in Section 3.2, the selection of parameters λ_1 and λ_2 influences the sparsity pattern of α_s. Based the Cholesky-based implementation of the LARS-Lasso algorithm, an homotopy method [66,72] can obtain the solutions for all possible values of regularization values. For the elastic net problem, similar procedure can be adopted while the elastic net can be transformed to a lasso problem as shown in Eq. (11). For each fixed λ_2, LARS is used to produce the entire solution path of the elastic net, then the chosen λ_2 is the one giving the highest reconstruction accuracy. The step k that arrives at the highest accuracy can be used as another parameter instead of λ_1. We refer the interested readers to the section 3.4 of literature [9] for details.

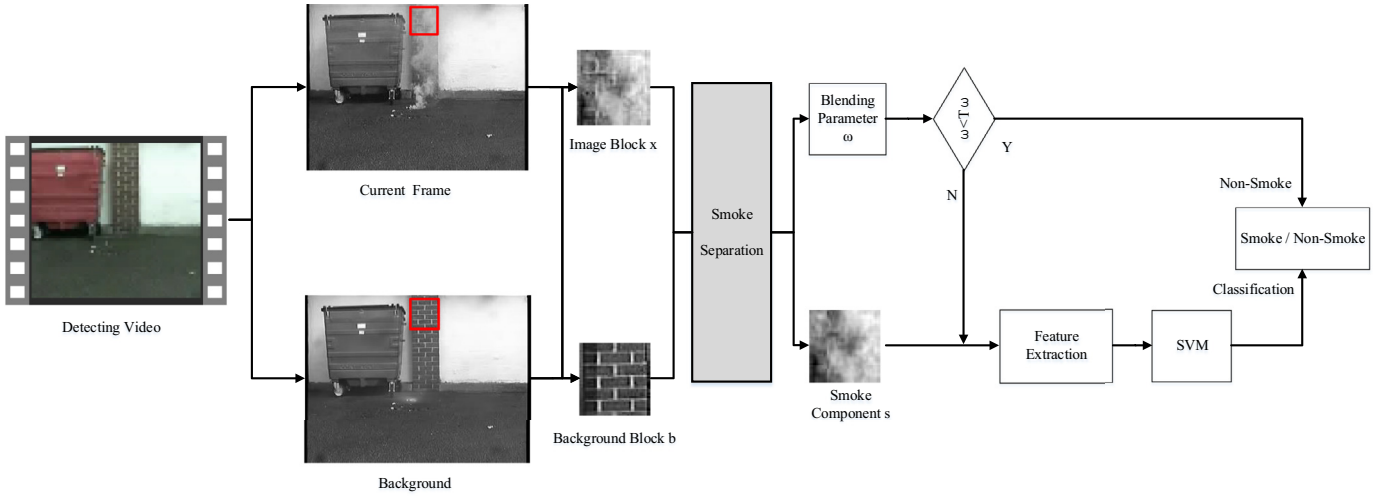

Fig. 2. Proposed video smoke detection framework.

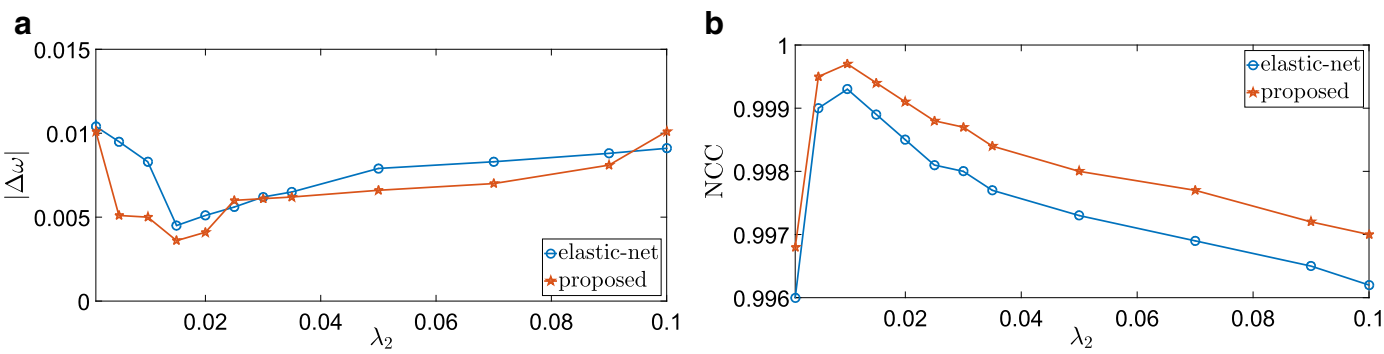

Fig. 3. Parameter choice of λ_2 for elastic-net.

We used a relatively small grid of values set for λ_2 to find the most suitable one. While we set $\lambda_2 \leqslant 1e-3$, it does not lead to convergence when the steps reach to the maximum value we set. When we set $\lambda_2 \geqslant 1e-1$, unsatisfactory results are obtained. We set blending parameter as 0.5 to create 1000 smoke images by combining 1000 pure smoke images with 1000 non-smoke images, the value sets $\{0.05 \pm 0.01k | k \in \{1, 2, 3, 4, 5\}\}$ are tested for parameter choice. As shown in Fig. 3, λ_2 at $1e-2$ achieves better results than that of low absolute differences $|\triangle\omega|$ and high NCC values. There id no systematic way to select the exact value, but the empirical value $\lambda_2 = 1e-2$ works well for all test cases of smoke separation and detection tasks in this paper.

5.2. Data sets

According to the steps of experiment, data sets for each step are prepared:

- **D1**-data sets for learning smoke dictionary D_s and non-smoke dictionary D_b: small size image blocks are collected in this article, such that pixels in the same block share the same blending parameter. On the other hand, very small blocks may provide insufficient features information. We choose the size of 16×16 according to [6,63]. There are 4000 pure smoke samples used for training the smoke dictionary, and each sample has 16×16 pixels, K-SVD [59] algorithm is adopted to train an over-complete dictionary with 500 bases. Similarly, 4000 non-smoke images from the CIFAR-100 [73] are collected as the negative training samples. Our research is mainly focused on the gray scale smoke separation. Fig. 4 displays 50 basis samples out of the 500 bases from the learned over-complete smoke dictionary in Fig. 4(a) and non-smoke dictionary in Fig. 4(b) respectively.
- **D2**-synthesized smoke images with different blending parameters: according to the experiments requirements for assessing the separation performance on synthesized images, pure smoke images and non-smoke images are combined to make smoke covered images with different blending parameters. Fig. 5 shows the examples of synthesized smoke images, we set the non-smoke images as backgrounds, ten different blending parameters are set as 0.05, 0.10,..., 0.95, 1. Separation performance evaluation are carried out by comparing the separated smoke components and blending parameter with the original pure smoke and preset blending parameters. Four groups of smoke images are prepared with different blending parameters increasing from 0 to 1 with precision 0.1, 0.05, 0.02, 0.01
- **D3**-data sets of real smoke and non-smoke videos: Videos for background modeling, smoke separation and detection are available from the public smoke clips[1] and the ViSOR repository[2] [74]. The smoke clips in these data sets cover indoor and outdoor spaces with different illuminations, short or long distance surveillance scenes.
- **D4**-synthesized data sets for smoke classification using different sparse representation based methods: To validate the importance of smoke separation, five groups training/test

[1] http://www.signal.ee.bilkent.edu.tr/VisiFire/Demo/SmokeClips.
[2] http://www.openvisor.org.

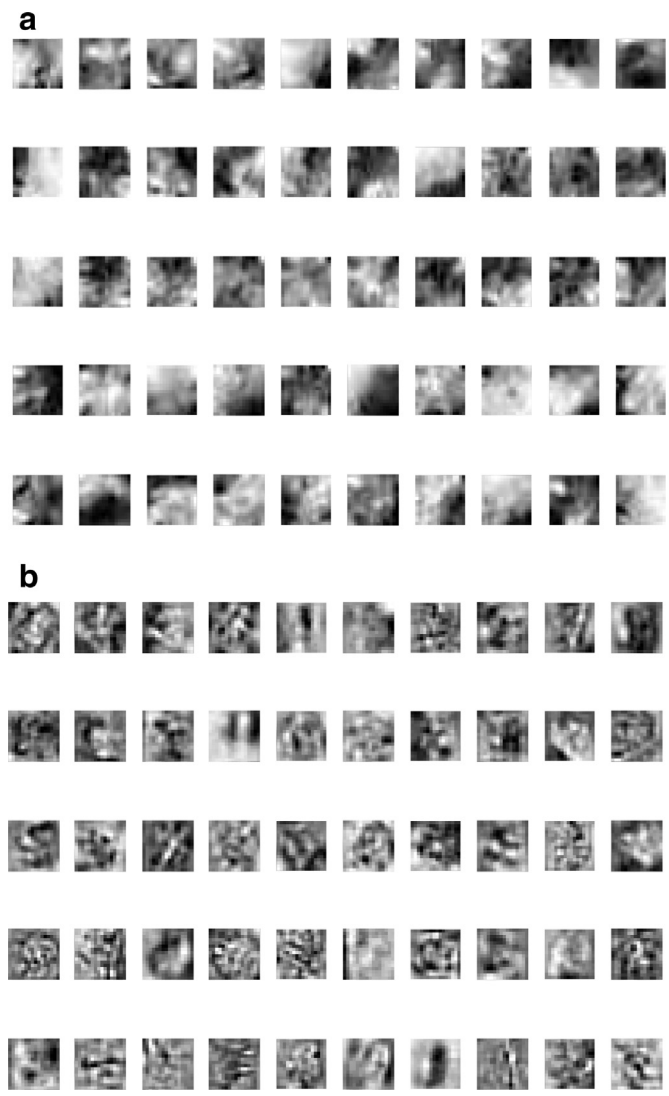

Fig. 4. (a) Basis samples from learned smoke dictionary D_s; (b) Basis samples from learned non-smoke dictionary D_b.

samples are produced with pure smoke and non-smoke samples from data sets D1. The five groups are different from each other due to the smoke image samples from light to heavy with five different blending parameters: 0.2, 0.4, 0.6, 0.8, 1.

- **D5**-data sets for training smoke classifier: We use pure smoke and non-smoke image sets from data sets D1 to create natural smoke images by combined smoke with non-smoke images, the non-smoke images are regarded as background b for smoke separation. Because the precision of blending parameters calculated is set as 0.01, we randomly set the blending parameters in the range of [0.01, 1]. 2000 smoke covered image with random blending parameters are produced for positive training samples and 2000 non-smoke images are selected as negative training samples. Similarly, other 2000 positive and negative image samples are collected for testing. Smoke detection on real videos is conducted on videos from D3.

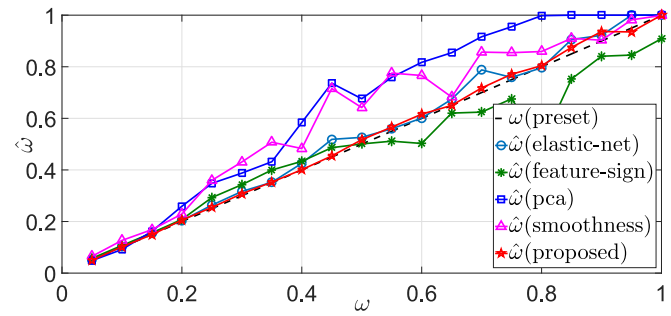

Fig. 6. Evaluation of smoke separation performance based on estimated blending parameters.

5.3. Smoke separation performance evaluation

The assessment criteria for separation performance can be divided into two parts. One is based on the difference between the estimated variables and the preset ones, which involve blending parameter ω and smoke component s. These two variables are unknown in real videos and we use the synthesized images from data sets $D2$. Another one is based on the visual evaluation with the real smoke or non-smoke videos $D3$. We compare our method with three other video smoke separation models [7]: feature-sign based sparse representation model, PCA based model and smoothness based model.

5.3.1. Smoke separation on synthesized images

Given a non-smoke image as background b and smoke image s, a set of blending parameters ω that increase in different precisions are adopted. The synthesized image can be represented as $x = \omega s + (1-\omega)b$. Separation algorithm is performed to estimate blending parameter $\widehat{\omega}$ and smoke component $\widehat{s}$.

We test four groups images with N samples (N=10, 20, 50, 100) respectively with ω of 0.1, 0.05, 0.02, 0.01. Fig. 6 shows the estimated blending parameters $\widehat{\omega}$. The two elastic-net based methods can achieve better results. It is noteworthy that the other three models achieve worse results of the estimated blending parameters $\widehat{\omega}$ while the elastic-net based methods hold stable separation performance based on the assessment of blending parameters. All groups of average absolute differences between the estimated $\widehat{\omega}$ and the preset ω are listed in Table 1. It presents smaller absolute differences for elastic-net spare representation method and the proposed method.

The normalized cross correlations algorithm (NCC) [75] is widely used for pattern matching and target tracking. In this paper, we apply NCC to evaluate the separation performance based on the smoke components estimation. It aims at comparing the matching degree between pure smoke samples s and the reconstructed

Fig. 5. Examples of synthesized smoke images with different blending parameters such as 0.05, 0.10,..., 0.95, 1.

Table 1
Average NCC and average absolute differences between the estimated $\widehat{\omega}$ and the preset ω for different combinations of the background and the pure smoke using different separation methods.

Change precision of ω	Local smooth	pca	Feature-sign	Elastic-net	Proposed
0.10	0.1522	0.1257	0.0600	0.0090	0.0083
0.05	0.1503	0.1022	0.0532	0.0088	0.0077
0.02	0.1571	0.1120	0.0532	0.0088	0.0063
0.01	0.1416	0.0977	0.0526	0.0086	0.0076
Average NCC	0.9633	0.9517	0.9865	0.9956	0.9991

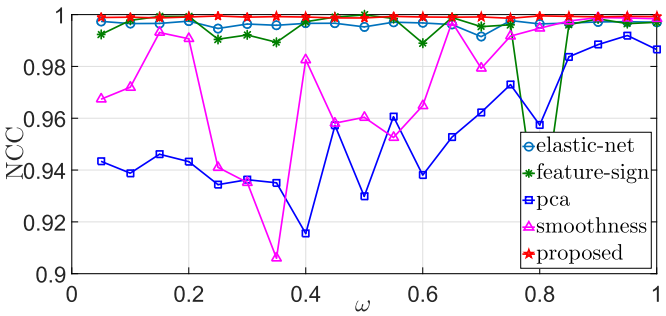

Fig. 7. Evaluation of smoke separation performance based on estimated en the separated smoke components and the preset smoke components.

smoke components $\widehat{s}$. Fig. 7 depicts the NCC calculated with the estimated and the true smoke components s by varying blending parameters. A higher NCC value usually indicates better separation performance. Better performance of the elastic-net based methods in estimation of average NCC is demonstrated in Table 1. From Fig. 7, separation by sparse representation shows a better performance. Our proposed method has led to the best and the most stable results among all the separation methods. From the performance of ω estimate and *NCC* values depicted in Figs. 6 and 7, the sparse representation based models for smoke separation outperform the others. Additionally, elastic-net based sparse coding can achieve more accurate results and the discriminative constraint of the proposed method can further improve the performance of elastic-net representation in smoke separation.

To further validate the improvements of the proposed method with discrimination part, we conduct the separation experiments on heavy smoke and non-smoke samples.

While the smoke detection performance depends highly on the separated smoke components, in order to see whether there exist the separated smoke components s or not, further experiment is conducted by using non-smoke image as x without any foregrounds covered. If such an s exists, it is under determined whether it looks like smoke or not. Fig. 8 shows the results on 20 image blocks. To show the estimated blending parameters, we set the original ω as non-zero but a very small value, say $1e-8$. As shown in Fig. 8, the estimated ω is nearly zero with well separated smoke components preformed on high NCC values. On the other hand, when the opaque objects are used as foregrounds with known background, the question arises as to the nature of the separated "smoke components" when x is a non-smoke image. Fig. 9 shows that the estimated ω values are all nearly 1.0 in these separations. Obviously, both methods can achieve good separation results while the proposed method has a small improvements. Accordingly, images in these two cases should be classified as non-smoke, while we use the over-complete smoke dictionary D_s to construct non-smoke images, the non-zero atoms in the sparse coefficients should be penalized by non-smoke dictionary D_b, which means they cannot encode non-smoke information over D_b simultaneously.

It is noted that while the nearly zero ω value can filter out the non-smoke image without any classification, difficulties occur when a scene is covered by the heavy smoke, the blending parameter will be 1.0 too. Thus further classification on separated foreground is needed while simply thresholding the blending parameter will not be sufficient in smoke detection tasks.

5.3.2. Smoke separation on real video frames

To test the proposed smoke separation method performed on real videos with uncertain smoke components and blending parameters, we display the separated images and give a visual assessment to the performance. Fig. 10 shows some separated smoke components on block level along with the estimated blending parameters. The separation performance indicates that the proposed method can achieve satisfactory results in real smoke videos with both texture (columns 1,2,6,7,10) and piece wise smooth backgrounds (columns 3,4,5,8,9). Note that the blending parameters are large (close to 1) when the background cannot be seen in the source image (column 2 and 6). Furthermore, Fig. 11 shows a few entire performances of the separated smoke components from our test videos. Scenarios with different challenges, short and long distance, heavy and light smoke, texture background and different illumination, are tested here. All the methods reported recently for smoke separation are used for comparison. To further evaluate the performance quantificationally, we calculate the objective function errors defined as

$$err = \frac{\|x-(1-\omega)b-\omega s\|_2}{\|x\|_2}. \tag{22}$$

As depicted in Fig. 11, the method using sparse representation can achieve better smoke separation results in visual assessment and quantitative evaluation. Smoke components can be well represented with sparse coefficients over learned smoke dictionary. Additionally, the values of objective function errors indicate that elastic-net sparse representation outperforms to feature-sign sparse representation in smoke separation.

5.4. Smoke detection performance evaluation

5.4.1. Smoke classification on synthesized data sets

Recently, several sparse representation over dictionary learning methods have been reported to be effective for image classification. Generally, the image samples are directly trained and tested for these methods without estimation of smoke components and blending parameters. Experiments are conducted here to verify the effectiveness of the proposed smoke detection method. It has been shown that learning a dictionary from the training samples instead of using all of them as a dictionary can further enhance the performance of SRC. Most existing classification-oriented dictionary learning methods try to learn discriminative class-specific dictionaries by either imposing block-diagonal constraints encouraging the incoherence between class-specific dictionaries [76]. We compare the proposed method with five methods: Sparse Representation-based Classification (SRC) [77], Kernel Sparse Representation (KSR) [78,79], Label Consistent K-SVD (LCKSVD) [61], Dictionary Learning with Structured Incoherence and Shared

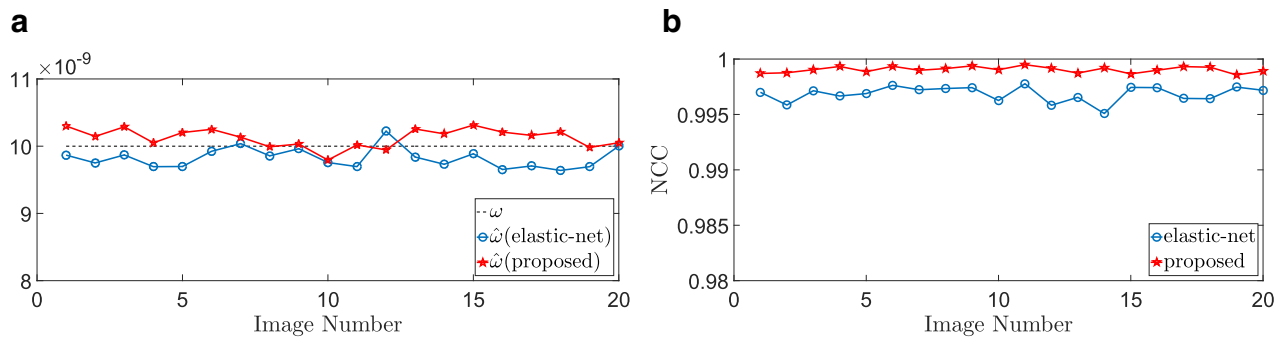

Fig. 8. Smoke separation on non-smoke image without foregrounds.

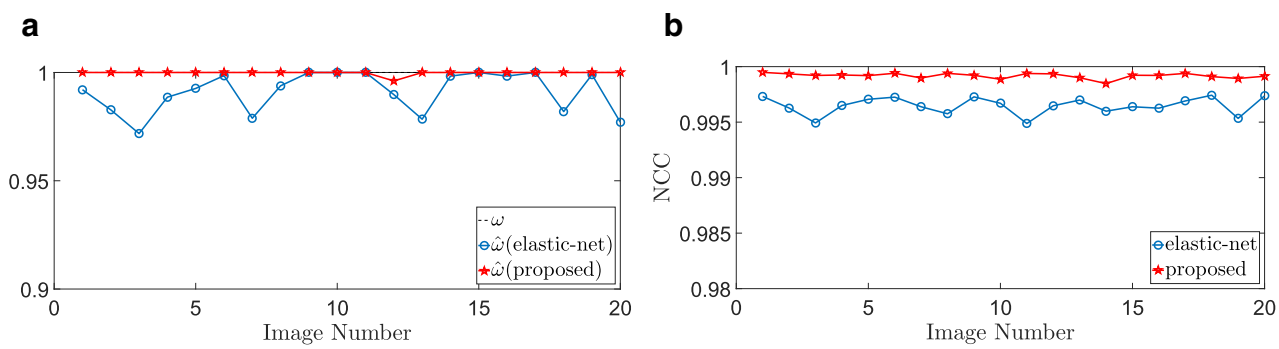

Fig. 9. Smoke separation on non-smoke images with opaque foregrounds.

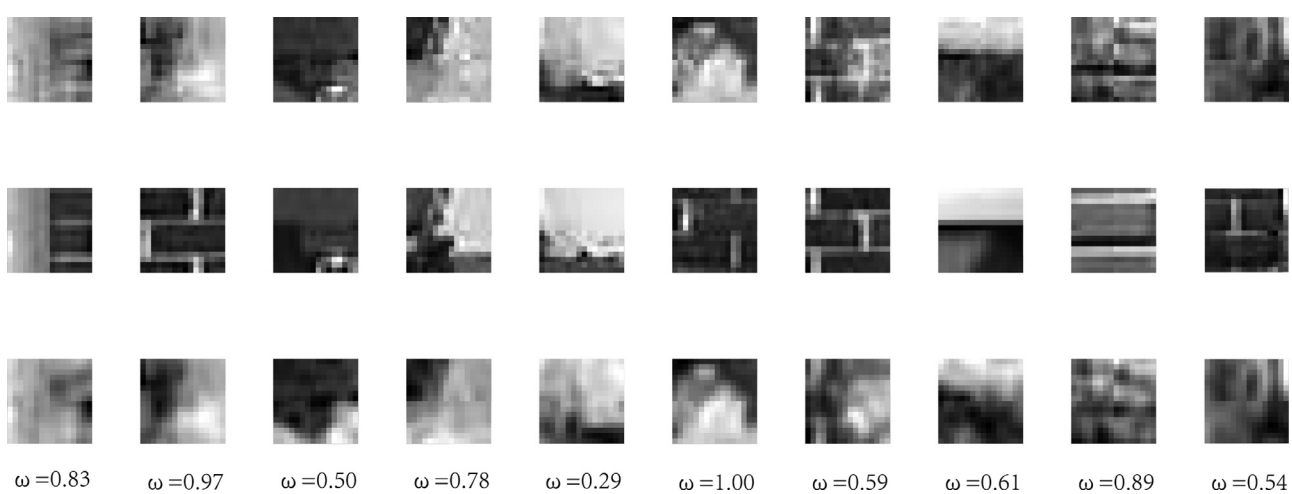

Fig. 10. Some blocks of separated smoke components from real smoke video frames. row 1: blocks in current input frame x, row 2: blocks in backgrounds b, row 3: blocks in separated smoke s.

Features (DLSI) [80], Fisher Discrimination Dictionary Learning (FDDL) [60], Dictionary Learning for Separating The Particularity–Commonality (COPAR) [81] and Low-rank Shared Dictionary Learning (LRSDL) [76].

The data set $D2$ is used for training and testing samples. We test 5 different groups of images with different blending parameters (0.2, 0.4, 0.6, 0.8, 1) where the smoke changes from light to heavy. For the classification methods without smoke separation, we treat the smoke covered images as positive training/test samples and non-smoke images as negative samples. The classification algorithm is referred to FDDL. For the smoke separation based classification, let x^s and x^b denote the pixels from smoke and non-smoke images respectively, we extract the sparse coefficients vector as feature as follows:

$$\alpha_s^s = \arg\min_{\alpha_s^s} \|x^s - (1-\omega)b - \omega D_s \alpha_s^s\|_2^2 + \Re(\alpha_s^s), \tag{23}$$

$$\alpha_s^b = \arg\min_{\alpha_s^b} \left\|x^b - D_s \alpha_s^b\right\|_2^2 + \Re\left(\alpha_s^b\right), \tag{24}$$

where α_s^s and α_s^b are sparse coefficients extracted from smoke and non-smoke components. $\Re(\alpha_s^c)(c = s, b)$ denotes the constraints of sparse representation:

(1) feature-sign: $\Re(\alpha_s^c) = \lambda\|\alpha_s^c\|_1$.
(2) elastic-net: $\Re(\alpha_s^c) = \lambda_1\|\alpha_s^c\|_1 + \lambda_2\|\alpha_s^c\|_2^2$.
(3) proposed: $\Re(\alpha_s^c) = \|D_b\alpha_s^c\|_2^2 + \lambda_1\|\alpha_s^c\|_1 + \lambda_2\|\alpha_s^c\|_2^2$.

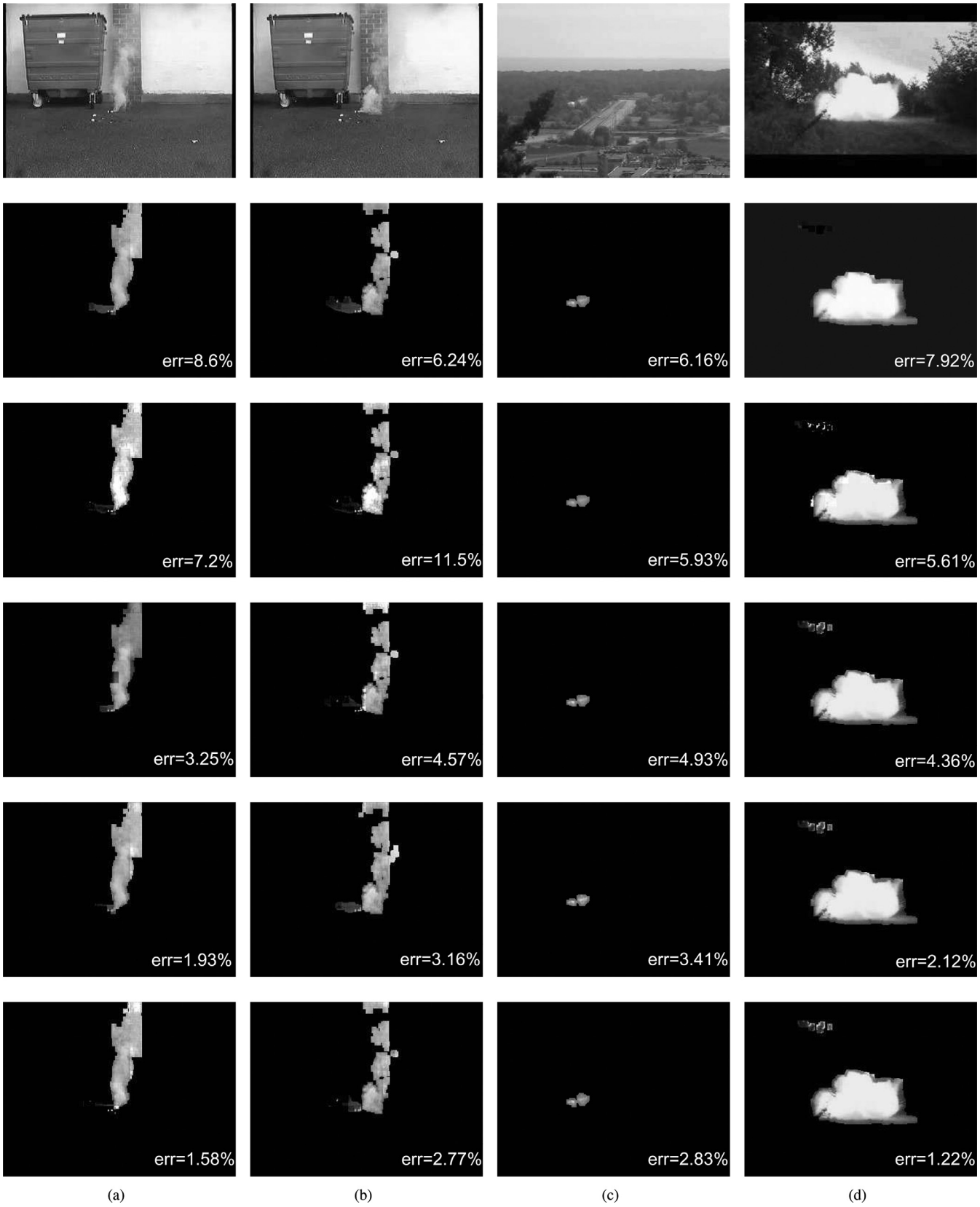

Fig. 11. Smoke separation from real videos using different methods. first row: the current input frame; second row: local smooth method; third row: pca method; fourth row: feature-sign method; fifth row: elastic-net method; sixth row: proposed method.

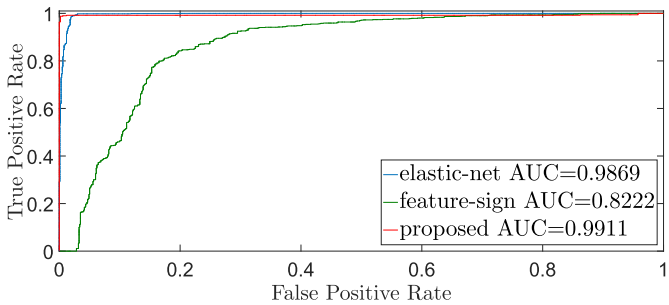

Fig. 12. ROC curves for classification of smoke and non-smoke on real videos, sparse coefficients are used as features.

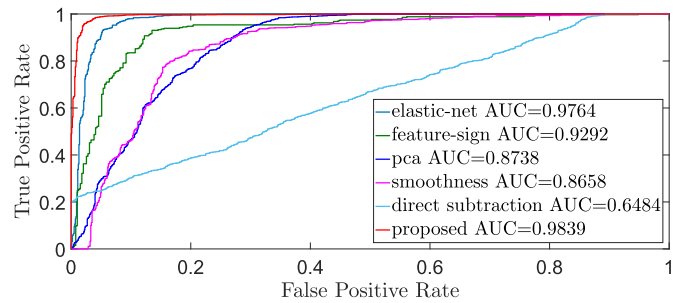

Fig. 13. ROC curves for classification of smoke and non-smoke on real video, LBP based features are used for classification.

SVM is adopted as the classifier for training and testing the sparse coefficients feature. The experimental results are listed in Table 2. For classification methods without smoke separation, the classification performs poorer on light smoke than that on heavy smoke. It indicates that the visible background can impact the feature extraction to some extent. Relatively, it has less impacts on visible background for those methods with smoke separation. Note that the feature-sign based method cannot work well in classification like elastic-net based methods [7]. This experiment demonstrates that the separation of smoke is important in smoke detection and our proposed method can extract the most discriminative features for smoke detection.

5.4.2. Smoke detection on real video sequences

In this section, we focus on performance of smoke detection methods with smoke separation function. According to the framework depicted in Fig. 2, after video smoke separation, we set $T_\omega = 0.01$, in the case $\omega \geqslant T_\omega$, features can be extracted from the separated smoke components.

Sparse coefficients (*SC*) and LBP features extracted from the smoke component (LBP_S) are used for classification respectively. We compare the proposed method with other three methods in [7]. In evaluating the LBP feature extracted from separated smoke components via different methods, we set the direct background subtraction method (ROSL [5]) as the baseline to validate that it is a simple method and is not appreciated in smoke separation.

The *SC* features can be obtained using the methods described in Section 5.3.1 with respect to smoke sparse coefficients α_s^s and non-smoke sparse coefficients α_s^b. Fig. 12 shows the ROC curves with respect to *SC* features extracted via the feature-sign search, the elastic-net method and the proposed method. Fig. 13 shows the LBP features extracted from the separated smoke components among all the methods. All corresponding classification accuracy are listed in Table 3. As expected, the texture features LBP extracted with these methods to the point of smoke separation obtained much better performance than that extracted by direct background subtraction method. Furthermore, the proposed method provides more reliable smoke information for LBP feature extraction. It is noteworthy that the detection performance of direct subtraction achieves only nearly 50% accuracy. It indicates that the extracted features do not have discriminative power due to the deteriorates from visible background in smoke image. While we expect that the sparse coefficient vector can encode reliable information of smoke, the coefficients vector gives us unsatisfactory results [63]. The proposed method using elastic net can figure out more robust and reliable feature vectors in sparse coding to characterize smoke components as shown in Fig. 12 and Table 3. Fig. 14 depicts the detection results on several real videos using the proposed *SC* features (the pictures in color version are used for better presentation). Fig. 14(a) shows the current source sequence, for each video, moving objects are located by simple motion detection, each block in motion regions is classified, detection is performed on the overlapped blocks. As shown in Fig. 14(b), blocks classified as heavy smoke ($\omega \in [0.6, 1]$), light smoke ($\omega \in [0.01, 0.6)$) and non-smoke are marked by red, yellow and green rectangles respectively. As shown in this figure, although there exist a few misclassification, the selected regions marked by all rectangles will not be misclassified if a simple majority voting is employed in each motion region.

Table 3
Classification accuracies(%) using LBP feature extracted from separated smoke components *LPBs* and sparse coefficients *SC*.

Feature	LBP_S	*SC*
Direct subtraction	54.57	/
Smoothness	81.60	/
PCA	83.19	/
Feature-Sign	88.60	79.41
elastic-net	90.00	96.31
Proposed	92.17	98.04

5.5. Computational complexity

There exist three main steps in the proposed smoke detection algorithm: motion detection of entire video, separating smoke components and detecting smoke block by block. Computationally, the proposed smoke separation spends most computation on obtaining the sparse codes to separate the smoke components, which is the solution to Eq. (18) with respect to blending parameter ω and sparse coefficients α_s. As described in Section 3.2, the proposed algorithm can be solved using lasso solution with tunning

Table 2
Classification accuracy(%) using different sparse representation methods.

ω	SRC	KSR	LCKSVD	DLSI	FDDL	COPAR	LRSDL	Feature-sign	Elastic-net	Proposed
0.2	59.58	60.55	62.95	60.85	62.90	60.68	60.75	69.45	85.71	89.15
0.4	64.05	62.17	69.87	69.17	71.70	67.02	67.78	70.17	88.57	90.86
0.6	69.40	77.11	73.70	72.02	74.37	71.05	71.02	73.22	93.67	93.76
0.8	79.75	80.00	82.92	81.08	82.70	80.75	81.35	71.00	94.29	96.95
1.0	87.50	89.12	84.62	93.25	89.70	93.00	84.15	73.79	92.15	96.25

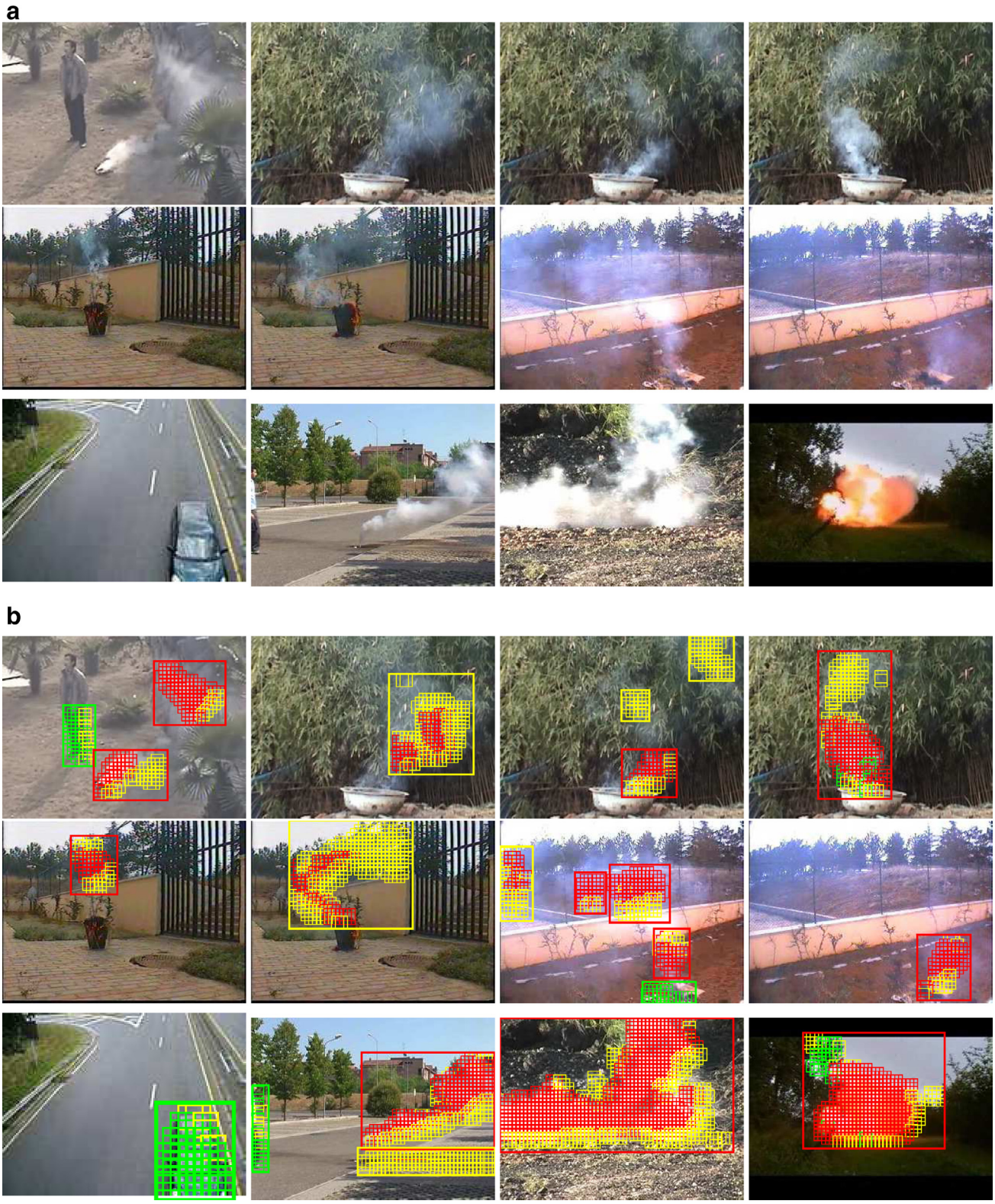

Fig. 14. (a) Examples of current input video frames; (b) Corresponding smoke detection results.

the parameter λ_2. Refer to [66], the theoretical computational cost of the algorithm is $\mathcal{O}(\aleph_1\aleph_2 L^2 m)$, where $\aleph_1$ is the number of points the λ_2 grid has, $\aleph_2$ is the number of data splits used in cross-validation, L, gives the number of features in the sparse coefficients α_s and m is the sample size. In this paper, the values of $\aleph_1$, $\aleph_2$, L and m are set as 5, 10, 500 and 256 respectively. We implement the separation algorithm in MATLAB 2016a on a PC with 3.40 GHZ Intel(R) Core(TM) i5 CPU and 8G memory. The run time of separation on each image block is about 0.025–0.042 s and about 0.014 s for detection. The entire time spent on each frame depends on the motion area of it. Notice that each block can be processed independently so that it is highly parallel.

6. Conclusions

Since a smoke image is linearly blended with smoke and background components, its pixels belonging to a small block are expected to share the same blending parameter. Features extracted from separated smoke components are deemed to encode more reliable information for smoke detection. In this article, a novel method is proposed for smoke separation, using the constraints of sparse representation norm, an optimization scheme for estimating the blending parameters and separating smoke components is developed based on elastic net. It combines the advantages of lasso and ridge regression, an overcomplete smoke dictionary D_s is used to reconstruct the smoke components s with the corresponding coefficients α_s as $s = D_s\alpha_s$. Improvements are obtained based on the discrimination rule that non-smoke dictionary is used to distinguish the atoms which contribute to the construction of smoke over learned smoke dictionary. Two types of features, coefficient vector α_s used directly as feature vector and LBP feature extracted from separated smoke components, are tested to verify the effectiveness of the proposed method. Experiments of smoke separation on synthesized images and real videos indicate that the proposed method can separate the smoke components in a more robust way from both quantitative analysis and visual assessments. Smoke detection on real videos gives the extensive proof that the proposed method for smoke separation and feature extraction significantly outperforms the existing methods.

Disclosure of conflicts of interest

None.

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